**Course: Innovative Data Management (M2 2IS)**

# Sales Behavior Analysis: Bridging Web Logs and Retail Transaction Data

Date: March 2026

Authors: Ibrahim Elhout, Ahmad Ibrahim, Mintesnot Yimer

M2 MIAGE Innovative Information Systems 2025–2026

**Table of Content**


| | |
|---|---|
| **Abstract** | **2** |
| **1. Introduction & Analysis Goals** | **2** |
| **2. Data Sources & Environment** | **2** |
| **3. Data Preparation Process** | **3** |
| **4. Implementation & Technical Justification** | **4** |
| **5. Results & Sales Behavior Comparison** | **5** |
| **6. Limitations & Methodological Considerations** | **8** |
| **7. Conclusion & Strategic Recommendations** | **9** |
| **References** | **12** |


**<u>List of Tables</u>**


**Table 1 — Category distribution across sales channels**

**Table 2 —Product revenue distribution by channel presence**

**Table 3—Department efficiency: web traffic vs revenue generation**

**Table 4—Web sales funnel progression and conversion rates**

**Table 5 — Price tier impact on conversion friction**

## Abstract

This report presents a crosschannel retail analytics pipeline comparing customer browsing behavior captured in Apache HTTP access logs against confirmed transactions from a structured retail database. Using a polyglot processing stack, Java MapReduce for log ETL, Scala for functional data transformation, and Spark SQL for complex relational analytics, we execute seven targeted analyses covering category presence, product ranking mismatches, department efficiency, conversion funnels, price sensitivity, and temporal purchasing patterns. Results reveal that while 91.43% of revenue-generating categories have a web presence, 60% of the full product catalog is visible online, and critical departments such as Fitness and Outdoors show severe conversion problems with gaps of -15.52 and -13.93 percentage points, respectively, between traffic and revenue generation. Actionable recommendations are proposed to address catalog gaps, conversion blockages, and campaign timing optimization.

## 1. Introduction & Analysis Goals

Modern retail environments are increasingly omnichannel, making it critical to understand how customer interactions online (browsing, adding to cart) translate to actual finalized purchases in a physical or direct-sale context. The core objective of this project is to perform a comparative analysis of sales behavior by bridging unstructured web access logs with a structured relational retail database.

To achieve a comprehensive understanding of channel performance, we defined and executed seven targeted analytical goals:

1. **Category Presence & Catalog Coverage:** Do web visitors browse the same categories that generate store revenue, and what percentage of the total catalog is visible online?
2. **Category Sales Proportions:** Are the proportions of sales across product categories similar across both channels?
3. **Product Ranking & Gaps:** Which specific products attract high web interest but fail to convert to actual sales?
4. **Department Efficiency:** Which departments generate high web traffic but low revenue (indicating severe conversion problems)?
5. **Sales Funnel Analysis:** Where do customers abandon the purchasing journey (Product View → Cart → Checkout → Add)?
6. **Price Influence:** How do different price tiers ($0-$50, $200+, etc.) impact conversion rates and channel performance?
7. **Temporal/Peak Hour Analysis:** When do customers browse online versus when they place actual orders?

## 2. Data Sources & Environment

This analysis synthesizes two highly divergent datasets, requiring significant data wrangling to align:

### Dataset 1: Structured Retail Transaction Database

Sourced from Cloudera, this represents the "source of truth" for confirmed in-store/direct sales. It consists of six hierarchical relational tables:

- **departments** (8 rows)
- **categories** (58 rows)
- **products** (1,345 rows)

- **customers** (12,435 rows)
- **orders** (68,883 rows)
- **order_items** (172,198 rows)

**Dataset 2: Unstructured Web Access Logs**

Representing online traffic, this dataset is an Apache HTTP daemon log file (access.log, approximately 40MB). It captures individual web requests (browsing behavior) but inherently lacks confirmed purchase data or quantity metrics, relying purely on IP addresses as a proxy for user sessions.

**Infrastructure & Environment Setup**

The project was deployed using a distributed Hadoop and Spark ecosystem containerized via Docker. The following commands initialize the infrastructure and execute the processing pipeline:

*Listing 1: Infrastructure setup and execution commands*

**1. Start the distributed Hadoop/Spark cluster**

```
docker-compose up -d
```

**2. Access the namenode container (master + Spark driver)**

```
docker exec -it namenode /bin/bash
```

**3. Create HDFS directories and upload raw datasets**

```
hdfs dfs -mkdir -p /input/weblog
hdfs dfs -mkdir -p /input/products
hdfs dfs put data/access.log /input/weblog/
hdfs dfs put data/.csv /input/
```

**4. Execute the MapReduce Data Preparation & Processing Jobs**

```
hadoop jar RetailProject.jar org.example. LogToCSV /input/weblog /input/clean_weblog
hadoop jar RetailProject.jar org.example. CategoryPresenceAnalysis /input/clean_weblog /input/
/output/mapreduce_analysis
```

**5. Execute the Spark/Scala Analytical Pipeline**

```
sparksubmit class org.example. spark. RetailAnalysis master local[] RetailProject.jar
```

## 3. Data Preparation Process

The raw access.log files are unstructured text (Combined Log Format) and cannot be natively joined with the relational database. A robust data preparation phase was implemented via Java MapReduce (LogToCSV.java) to parse the logs into a structured, queryable format.

We utilized a mapper-only MapReduce job (enforced via job.setNumReduceTasks(0)) to achieve highly parallelized extraction without the network overhead of the shuffle/sort phase.

**Extraction Methodology**

- **Regex Extraction:** Complex regular expressions (java.util.regex.Pattern) were deployed to dissect the HTTP requests. Four distinct patterns (P_FULL, P_CAT, P_DEPT, P_PAGE) were used to handle varying URL depths (e.g., /department/\{dept\}/category/\{cat\}/product/\{product_name\}/add_to_cart).

- **Filtering & Decoding:** Only successful requests (respCode == "200") were retained to filter out 404 errors. URLDecoder.decode was applied to sanitize spaces (e.g., %20) and special characters.

- **Signal Definition:** Suffixes in the URL were parsed to classify the pageType. Crucially, an /add_to_cart request was designated as our primary *web sales proxy,* as it represents the strongest available signal of purchase intent in the log.

- **Output:** The mapper emitted a cleaned, comma-separated string wrapped in NullWritable.get(), resulting in a structured CSV (ip, datetime, dept, category, product, pageType, respCode) written directly to HDFS.

## 4. Implementation & Technical Justification

To process the data and extract business value, three distinct distributed computing paradigms were utilized, demonstrating technical maturity in choosing the right tool for specific analytical workloads.

### 4.1 MapReduce in Java (Low-Level ETL & Global Aggregation)

**Implementation:** MapReduce was utilized for highly customized joining and counting tasks (CategoryPresenceAnalysis.java and ProductCoverageAnalysis.java). We implemented a two-mapper architecture (multiple inputs) to read the weblog and store data simultaneously.

**Technical Justification:** A standard reduced-side join for this data would cause massive network shuffling. To optimize this, we utilized the distributed cache to load the smaller products.csv and categories.csv into a memory HashMap during the mapper's setup() phase, enabling a lightning-fast map-side join. Furthermore, in the coverage analysis, we deliberately forced exactly one reducer (job.setNumReduceTasks(1)) to ensure that global counters correctly calculated the final sitewide catalog coverage percentages.

### 4.2 Scala (Functional Data Manipulation)

**Implementation:** Scala was used alongside the Spark DataFrame API (CategoryPresence.scala) as a proof of concept for functional data cleaning.

**Technical Justification:** Scala's functional syntax pairs perfectly with Spark's lazy evaluation. Using .withColumn() and native SQL functions like substring_index() and initcap() allowed for elegant, inline string manipulation to remove URL fragments, a process that is notoriously verbose and rigid in raw Java MapReduce.

### 4.3 DataFrames / Spark SQL (Complex Relational Analytics)

**Implementation:** The core analytical heavy lifting (RetailAnalysis.java) was executed via Spark SQL. Data was loaded from HDFS and registered as temporary SQL views (createOrReplaceTempView).

**Technical Justification:** Spark SQL was chosen because of its Catalyst optimizer, which mathematically determines the most efficient execution plan. This was strictly necessary for our analyses, which required massive FULL OUTER JOIN operations.

**Overcoming the Row Multiplication Bug:** During Analysis 6 (Price Influence), we encountered a severe Cartesian product bug where joining products directly to weblogs and orders inflated counts by orders of magnitude. This approach reflects a deliberate engineering tradeoff by restructuring our SQL to use Common Table Expressions (CTEs). By preaggregating web data (web_agg) and store data (store_agg) independently before the join, we ensured data integrity. We also utilized advanced window functions (SUM(COUNT()) OVER()) to compute percentage shares without requiring multiple subqueries.

## 5. Results & Sales Behavior Comparison

Executing the Spark SQL pipelines revealed profound discrepancies between how customers interact with the brand online versus how they spend their money.

### Analysis 1: Category Presence & Catalog Coverage

There is exceptional alignment at the top level: 91.43% of categories with confirmed sales have a web presence (measured as categories appearing in both web cart-adds and store sales). This indicates very strong consistency between online browsing and physical purchasing behavior.

Comparing the web traffic against the entire database catalog reveals 60% coverage. This means 22 categories exist in the system but receive absolutely zero web traffic (e.g., "bike & skate shop," "boating," "boys' apparel," and "football"). These categories represent significant untapped digital opportunity—products exist in inventory but are invisible to online customers. opportunity. Products exist in inventory but are invisible to online customers.

| Presence Type | Category Count | Percentage |
|---|---|---|
| Both Channels (Web + Store) | 32 | 91.43% |
| Web Only | 3 | 8.57% |
| Store Only | 0 | 0% |
| **Total Categories** | **35** | **100%** |

Table 1: Category distribution across sales channels

Note: Against the full product catalog (55 categories), 33 have both web & catalog presence, 2 are web-only, and 22 are catalog-only (no web traffic) → 60% coverage.

### Analysis 2: Category Sales Proportions

Channel dominance varies significantly across product categories. For categories appearing in both channels, we measured the proportion each represents of total channel activity:

- **Trade-in:** Represents 2.21% of web activity but only 0.72% of store sales, a 1.49 percentage point gap. Customers browse trade-in options online but rarely complete transactions, suggesting trust barriers or unclear valuation processes.
- **Boxing & MMA:** Shows 0.73% web traffic but 0% store sales. This indicates pure research behavior without conversion, possibly due to pricing concerns or competitor availability.

- **Accessories:** Similar pattern with 0.60% web traffic but 0% store sales, suggesting customers discover accessories online but purchase elsewhere due to price sensitivity.

These imbalances highlight categories requiring targeted conversion optimization through clearer value propositions, enhanced product imagery, and trust-building elements such as customer reviews.

**Analysis 3: Product Ranking & Revenue Distribution**

We classified all 1,074 unique products into three groups based on their presence in web cart adds versus confirmed store sales:

| Presence Category | Product Count | Revenue Generated |
|---|---|---|
| Both Channels | 75 | $5.08M (~74%) |
| Store Only | 27 | $1.78M (~26%) |
| Web Only | 2 | $0 (0%) |
| **Total** | **104** | **~$6.86 million** |

Table 2: Product revenue distribution by channel presence

**Critical Finding:** There are 2 "Web Only" products that represent severe conversion failures: the "Nike men's Free 5.0 running shoe" (web rank #8, 374 cart adds) and the "Adidas kids' RG III mid football cleat" (web rank #2, 710 cart adds). Customers demonstrate clear purchase intent (added items to cart), but these products generate zero confirmed revenue. Possible root causes include:

1. Checkout barriers (payment processing issues, shipping restrictions, complex checkout flows)
2. Inventory synchronization problems (products appear available online but are out of stock)
3. Price transparency issues (final checkout price differs from displayed price)
4. Abandoned cart recovery failure (no email reminders or incentives)

**Specific Examples:**

- The "perfect fitness perfect rip deck" ranks #1 across both channels, demonstrating strong cross-channel appeal
- The "adidas kids' rg iii mid football cleat" shows high web interest (Web Rank #2, 710 cart adds) but generated zero store sales
- The "nike men's free 5.0 running shoe" received 374 web cart additions (Web Rank #8) but generated zero store sales
- Conversely, the "nike men's free 5.0+ running shoe" (note the "+" indicating a different model) sold 36,680 units in stores (Store Rank #4) but received zero web cart additions

**Analysis 4: Department Efficiency (Traffic vs Revenue Gap)**

For each department, we calculated the gap between its share of web traffic and its share of total revenue. A positive gap indicates efficiency (converts better than traffic would predict), while a negative gap indicates underperformance.

| Department | Web Traffic % | Revenue % | Gap |
|---|---|---|---|
| Fan Shop | 16.90% | 49.84% | +32.94 |
| Apparel | 16.69% | 21.18% | +4.64 |
| Golf | 16.60% | 13.43% | -3.17 |
| Footwear | 16.64% | 11.67% | -4.96 |
| Outdoors | 16.83% | 2.90% | -13.93 |
| Fitness | 16.34% | 16.34% | -15.52 |

Table 3: Department efficiency: web traffic vs revenue generation

**Critical Finding:** Fitness shows a -15.52 percentage point gap, the worst performance across all departments. Despite commanding 16.34% of web traffic, it generates only 0.82% of revenue. Outdoors follows with a -13.93 gap (16.83% traffic and 2.90% revenue). These massive underperformances suggest the following:

- Size and fit uncertainty driving cart abandonment
- Inadequate product photography or descriptions
- Noncompetitive pricing versus fast-fashion retailers
- Lack of styling suggestions or outfit inspiration

Conversely, Fan Shop demonstrates exceptional efficiency with a +32.94 gap, generating 49.84% of revenue from only 16.90% of traffic. This indicates extremely strong product-market fit, likely driven by licensed merchandise with high margins and emotional purchasing drivers. Apparel also shows healthy efficiency (+4.64), contrary to initial concerns.

**Analysis 5: Web Sales Funnel & Conversion Rates**

We tracked user progression through the web purchase funnel. Note that the web logs do not contain complete checkout flow data, so we focus on the cart-add behavior as the primary conversion signal.

| Funnel Stage | Event Count | Conversion Rate |
|---|---|---|
| Product Views | 30,519 | - |
| Add to Cart | 12,052 | 39.5% |

Table 4: Web sales funnel progression and conversion rates

**Critical Dropoff Point:** Key Insight: The 39.5% view-to-cart conversion rate is below typical e-commerce benchmarks (40-50% for sporting goods). This represents the primary friction point where nearly 60% of engaged browsers fail to demonstrate purchase intent.

Potential causes include the following:

- Price visibility issues or unexpected pricing
- Inadequate product information (specifications, sizing, materials)
- Lack of product reviews or social proof
- Complex navigation or add-to-cart process
- Comparison shopping behavior (checking competitor prices)
- Lack of trust signals or return policy clarity

**Analysis 6: Price Influence on Channel Performance**

We calculated "web events per unit sold" across price ranges to measure the research-to-purchase friction:

| Price Range | Average Price | Web Products | Cart Adds | Store Units Sold | Events per Unit Sold |
|---|---|---|---|---|---|
| $0 - 50 | $27.65 | 32 | 7,987 | 123,796 | 0.06 |
| $50$–$100$ | $79.78 | 15 | 4,637 | 181,347 | 0.03 |
| $100$ - $200 | $146.16 | 15 | 5,159 | 39,022 | 0.13 |
| $200+ | $413.69 | 12 | 2,602 | 31,593 | 0.08 |

Table 5: Price tier impact on conversion friction

**Key Insight:** The $50-$100 tier performs best with only 0.03 events per unit sold, indicating extremely efficient conversion. The $100-$200 tier underperforms significantly (0.13 events per unit), requiring 4.3x more engagement than the optimal $50-$100 tier. This suggests a "dead zone" where products are too expensive for impulse purchases but not premium enough to justify extensive research.

**Analysis 7: Temporal Patterns & Peak Hours**

Aggregating activity by hour of day revealed distinct behavioral patterns.

- **Web Browsing Peak:** 7:00 PM (evening leisure browsing)
- **Cart Addition Peak:** 7:00 PM (decision-making aligns with browsing)
- **Order Placement Peak:** 6:00 PM (actual purchases occur earlier)

**Strategic Implication:** Customers place confirmed orders at 6:00 PM but continue browsing at 7:00 PM. This suggests two distinct user segments:

1. Afterwork converters (6 PM) who make quick purchase decisions during or immediately after their commute
2. Evening researchers (7 PM) who gather information but may convert on subsequent visits

This temporal mismatch indicates that current email marketing campaigns likely target the 7 PM browsing window, missing the higher-conversion 5 PM purchasing window.

## 6. Limitations & Methodological Considerations

While this analysis provides significant business insights, several limitations should be acknowledged:

- **Web Sales Proxy:** The web dataset tracks add_to_cart events as purchase intent, but this does not represent confirmed transactions. Actual online sales may differ from cart-add behavior due to abandonment.
- **User Identification:** Web logs rely on IP addresses as user proxies. Multiple users may share IP addresses (workplace, household), and single users may appear as multiple IPs (mobile, home, work networks).
- **Temporal Scope:** The datasets represent a single day of activity (June 14, 2014), limiting our ability to detect seasonal patterns, promotional impacts, or long-term trends.
- **Channel Attribution:** The analysis assumes clean channel separation. In reality, customers may research online (web cart add) and then purchase in-store, creating attribution challenges not captured in this dataset linkage.

Despite these constraints, the cross-channel patterns identified remain valuable for strategic decision-making and operational optimization.

## 7. Conclusion & Strategic Recommendations

### Technical Takeaways

This project successfully demonstrated the necessity of polyglot persistence and processing in big data environments. While Hadoop MapReduce is unparalleled for distributed, schema-on-read parsing of raw log files, it is too rigid for complex analytics. Spark SQL, leveraging CTEs and the Catalyst Optimizer, proved vital in preventing Cartesian data inflation and efficiently executing the windowed aggregations required to compare these divergent datasets. Furthermore, standardizing timestamps using TO_TIMESTAMP() and text via LOWER(TRIM()) proved to be critical data engineering prerequisites for successful cross-system joins.

### Business Insights & Strategic Recommendations

Based on the seven analyses conducted, we propose the following prioritized action plan:

**Recommendation 1**: Close the Digital Catalog Gap (60% coverage)

**Finding:** 22 categories receive zero web traffic despite existing in the product catalog.

**Action:** Conduct comprehensive SEO audit, create dedicated category landing pages, optimize site navigation to surface missing categories, and implement internal linking strategies.

**Expected Impact:** 2030% increase in product discovery, estimated revenue expansion of $150K–$250K annually from previously unutilized inventory.

**Recommendation 2:** Address Web-Only Product Conversion Failure (2 critical products, \$0 revenue)

**Finding:** 2 high-traffic products generate significant cart-adds but zero confirmed sales: Nike Free 5.0 (374 cart adds, rank #8) and Adidas RG III Cleat (710 cart adds, rank #2).

**Action:**

- Conduct detailed checkout funnel analysis for these specific SKUs
- Verify inventory synchronization between web and POS systems
- Implement abandoned cart recovery email campaigns with personalized incentives
- Add product-specific trust signals (reviews, detailed specifications, return policy)

**Expected Impact:** Recovery of 1525% of abandoned cart value, estimated at $75K–$125K in incremental revenue.

**Recommendation 3**: Fix Fitness & Outdoors Conversion Crisis (-15.52 and -13.93 gaps)

**Finding:** Fitness receives 16.34% of traffic but generates only 0.82% of revenue (worst department). Outdoors receives 16.83% of traffic but generates only 2.90% of revenue (second-worst).

**Action:**

- Conduct urgent pricing audit against competitors (REI, Dick's Sporting Goods)
- Verify inventory synchronization; confirm web-listed items are actually in stock
- Add free shipping thresholds or flat-rate shipping for heavy fitness equipment
- Implement detailed product comparison tools for fitness equipment
- Add customer review systems with verified purchase badges
- Create educational content (buying guides, workout programs, setup videos)
- Implement financing options for premium equipment ($500+)

**Expected Impact:** 10-15 percentage point gap reduction, translating to \$300K-\$450K in incremental revenue from currently underperforming departments.

**Recommendation 4:** Optimize Cart Abandonment (49.49% drop-off)

**Finding:** Half of cart users fail to proceed to checkout, the single largest friction point in the conversion funnel.

**Action:**

- Simplify checkout process (reduce form fields, enable autofill)
- Implement guest checkout option
- Display total cost (including shipping) earlier in the funnel
- Add prominent trust signals (SSL badges, return policy, customer service contact)
- Offer multiple payment options (PayPal, Apple Pay, Afterpay)
- Deploy exit-intent popups with discount incentives

**Expected Impact:** 10.15 percentage point improvement in cart checkout conversion, estimated $300K–$500K annual revenue increase.

**Recommendation 5:** Realign Campaign Timing (6 PM purchase peak vs 7 PM browse peak)

**Finding:** Email campaigns likely target a 7 PM browsing window, but actual purchases peak at 6 PM.

**Action:** Shift promotional email send times to the 4:30-5:30 PM window to intercept the high-conversion after-work audience. Implement A/B testing to validate optimal send time by segment.

**Expected Impact:** 2030% improvement in email campaign conversion rates with zero additional cost, purely through timing optimization.

**Recommendation 6:** Premium Product Support ($200+ requires 3.9x more engagement)

**Finding:** High-value products require significantly more touchpoints before purchase.

**Action:**

- Create detailed comparison charts for premium products
- Provide financing options (installment payments, buy now pay later).
- Implement live chat support for high-ticket category pages
- Develop comprehensive buyer's guides and educational content
- Enable "Save for Later" and price drop alert features

**Expected Impact:** 812% improvement in premium product conversion and an estimated $100K–$175K incremental revenue.

**Summary**

The integration of unstructured web behavior logs with structured transactional data has revealed actionable insights across channel alignment, conversion optimization, department performance, and customer timing. By implementing these six prioritized recommendations, the organization can address catalog visibility gaps, reduce friction points, optimize resource allocation, and align marketing timing with actual purchasing behavior, collectively representing an estimated $825K to $1.4 million in annual incremental revenue opportunity.

## References

[1] Apache Software Foundation. (2024). *Apache Hadoop Documentation*. https://hadoop.apache.org/docs/

[2] Apache Software Foundation. (2024). *Apache Spark SQL Guide*. https://spark.apache.org/docs/latest/sqlprogrammingguide.html

[3] Cloudera, Inc. (2014). *Retail Database Sample Dataset*. Cloudera QuickStart VM Distribution.

[4] World Wide Web Consortium. (2024). *Combined Log Format Specification*. https://www.w3.org/Daemon/User/Config/Logging.html